

An Adaptive Universal Trust-Region Method with Additive Search Overhead

Abstract

We first give a refined analysis of the simple universal trust-region method that removes its multiplicative logarithmic factor. We then present an adaptive variant for finding approximate second-order stationary points without knowing the Lipschitz constant of the Hessian. The method uses a gradient-clipped regularization scale and a multiplier-aware acceptance test. A logarithmic potential controls gradient-contraction steps, while a persistent regularization parameter amortizes rejected trials. If the Hessian is M -Lipschitz continuous and $\rho_0 \leq c\sqrt{M}$ for a constant $c > 0$, the adaptive method returns a point $\hat{x}$ satisfying

$$\|\nabla f(\hat{x})\| \leq \epsilon, \quad \lambda_{\min}(\nabla^2 f(\hat{x})) \geq -O(\sqrt{M}\epsilon)$$

using

$$O\left(\sqrt{M}(f(x_0) - f^*)\epsilon^{-3/2} + \log_+(\|\nabla f(x_0)\|/\epsilon) + \log_+(\sqrt{M}/\rho_0)\right)$$

trial steps. Both logarithmic terms are additive.

1 Problem and assumptions

We consider

$$\min_{x \in \mathbb{R}^n} f(x), \tag{1.1}$$

where f is twice continuously differentiable and bounded below by $f^* > -\infty$. The construction follows the gradient-scaled trust-region framework of Jiang et al. [1].

Assumption 1.1. *There exists $M > 0$ such that*

$$\|\nabla^2 f(x) - \nabla^2 f(y)\| \leq M\|x - y\| \quad \forall x, y \in \mathbb{R}^n. \tag{1.2}$$

Under [Assumption 1.1](#), for all $x, d \in \mathbb{R}^n$,

$$\|\nabla f(x + d) - \nabla f(x) - \nabla^2 f(x)d\| \leq \frac{M}{2}\|d\|^2, \tag{1.3a}$$

$$f(x + d) - f(x) - \nabla f(x)^T d - \frac{1}{2}d^T \nabla^2 f(x)d \leq \frac{M}{6}\|d\|^3. \tag{1.3b}$$

For a prescribed $\epsilon > 0$, the target conditions are

$$\|\nabla f(x)\| \leq \epsilon, \quad \lambda_{\min}(\nabla^2 f(x)) \geq -O(\sqrt{M}\epsilon). \quad (1.4)$$

For $t > 0$, we use $\log_+(t) := \max\{\log t, 0\}$. Let $\mathbb{N}_0 := \{0, 1, 2, \dots\}$.

2 A refined count for the simple UTR method

Assume in this section that M is known. Let $a_k := \|\nabla f(x_k)\|$. The simple strategy of Jiang et al. [1] computes a global solution of

$$\begin{aligned} \min_{d \in \mathbb{R}^n} \quad & \nabla f(x_k)^T d + \frac{1}{2} d^T \left(\nabla^2 f(x_k) + \frac{\sqrt{M}}{3} \sqrt{a_k} I \right) d, \\ \text{s. t.} \quad & \|d\| \leq \frac{\sqrt{a_k}}{3\sqrt{M}}. \end{aligned} \quad (2.1)$$

Let λ_k be the trust-region multiplier. With the convention $\inf \emptyset = \infty$, define

$$T_\epsilon := \inf\{t \in \mathbb{N}_0 : a_t \leq \epsilon\} \in \mathbb{N}_0 \cup \{\infty\}. \quad (2.2)$$

Theorem 2.1 (Refined simple-UTR complexity). *Under [Assumption 1.1](#), the method (2.1) has $T_\epsilon < \infty$ and satisfies*

$$T_\epsilon \leq \left(81 + \frac{6}{\log 6}\right) \sqrt{M} (f(x_0) - f^*) \epsilon^{-3/2} + \frac{\log_+(a_0/\epsilon)}{\log 6} + 1. \quad (2.3)$$

Proof. Write $g_k = \nabla f(x_k)$, $H_k = \nabla^2 f(x_k)$, and $\Delta_f = f(x_0) - f^*$. We first derive the two estimates used below. This is the specialization of the function-decrease and gradient-norm estimates in Jiang et al. [1]; the details are included here to make the argument self-contained. Set

$$\sigma := \frac{\sqrt{M}}{3}, \quad R_k := \frac{\sqrt{a_k}}{3\sqrt{M}}.$$

For a global solution d_k of (2.1), the trust-region optimality conditions give a multiplier $\lambda_k \geq 0$ such that

$$\begin{aligned} \|d_k\| &\leq R_k, \quad \lambda_k(\|d_k\| - R_k) = 0, \\ (H_k + \sigma\sqrt{a_k}I + \lambda_k I)d_k &= -g_k, \quad H_k + \sigma\sqrt{a_k}I + \lambda_k I \succeq 0. \end{aligned}$$

Consequently, (1.3b), stationarity, and positive semidefiniteness imply

$$\begin{aligned} f(x_{k+1}) - f(x_k) &\leq g_k^T d_k + \frac{1}{2} d_k^T H_k d_k + \frac{M}{6} \|d_k\|^3 \\ &= -\frac{1}{2} d_k^T (H_k + \sigma\sqrt{a_k}I + \lambda_k I) d_k - \frac{1}{2} (\sigma\sqrt{a_k} + \lambda_k) \|d_k\|^2 + \frac{M}{6} \|d_k\|^3 \\ &\leq -\left(\frac{\sqrt{M}a_k}{6} - \frac{M}{6} \|d_k\|\right) \|d_k\|^2 - \frac{\lambda_k}{2} \|d_k\|^2. \end{aligned} \quad (2.4)$$

Using $\|d_k\| \leq R_k$ in the last line yields

$$f(x_k) - f(x_{k+1}) \geq \frac{\sqrt{Ma_k}}{9} \|d_k\|^2 + \frac{\lambda_k}{2} \|d_k\|^2 \geq 0. \quad (2.5)$$

Likewise, (1.3a), stationarity, and $\|d_k\| \leq R_k$ give

$$\begin{aligned} a_{k+1} &\leq \|\nabla f(x_{k+1}) - g_k - H_k d_k\| + \|g_k + H_k d_k\| \\ &\leq \frac{M}{2} \|d_k\|^2 + (\sigma\sqrt{a_k} + \lambda_k) \|d_k\| \\ &\leq \frac{a_k}{18} + \frac{a_k}{9} + \frac{\lambda_k \sqrt{a_k}}{3\sqrt{M}} = \frac{1}{6} a_k + \frac{\lambda_k \sqrt{a_k}}{3\sqrt{M}}. \end{aligned} \quad (2.6)$$

If $\lambda_k > 0$, complementarity gives $\|d_k\| = R_k$. Substitution into (2.5), together with (2.6), proves

$$f(x_k) - f(x_{k+1}) \geq \frac{a_k^{3/2}}{81\sqrt{M}} + \frac{\lambda_k a_k}{18M}, \quad (2.7a)$$

$$a_{k+1} \leq \frac{1}{6} a_k + \frac{\lambda_k \sqrt{a_k}}{3\sqrt{M}}. \quad (2.7b)$$

If $\lambda_k = 0$, (2.6) gives

$$a_{k+1} \leq \frac{1}{6} a_k. \quad (2.8)$$

Equation (2.5) also shows directly that every step is monotone in function value.

If $a_0 \leq \epsilon$, then $T_\epsilon = 0$. Hence assume $a_0 > \epsilon$. Fix any $N \in \mathbb{N}_0$ with $N < T_\epsilon$ and define

$$\begin{aligned} \mathcal{B}_N &:= \{0 \leq k < N : \lambda_k > 0\}, \quad \mathcal{I}_N := \{0 \leq k < N : \lambda_k = 0\}, \\ q_k &:= \frac{\lambda_k}{\sqrt{Ma_k}} \quad (k \in \mathcal{B}_N). \end{aligned}$$

Since $a_k > \epsilon$ for $0 \leq k \leq N$ and all steps are monotone, summing (2.7a) over $\mathcal{B}_N$ gives

$$|\mathcal{B}_N| \leq 81\sqrt{M} \Delta_f \epsilon^{-3/2}, \quad \sum_{k \in \mathcal{B}_N} q_k \leq 18\sqrt{M} \Delta_f \epsilon^{-3/2}. \quad (2.9)$$

Moreover, for $k \in \mathcal{B}_N$, (2.7b) implies

$$\left[\log \frac{a_{k+1}}{a_k} \right]_+ \leq \left[\log \left(\frac{1}{6} + \frac{q_k}{3} \right) \right]_+ \leq \frac{q_k}{3}. \quad (2.10)$$

Telescoping $\log a_k$ from 0 to N and using $a_N > \epsilon$ gives

$$\log \frac{a_N}{a_0} \leq -|\mathcal{I}_N| \log 6 + \frac{1}{3} \sum_{k \in \mathcal{B}_N} q_k,$$

and hence

$$|\mathcal{I}_N| \leq \frac{\log(a_0/\epsilon) + \frac{1}{3} \sum_{k \in \mathcal{B}_N} q_k}{\log 6}. \quad (2.11)$$

Consequently, every $N \in \mathbb{N}_0$ with $N < T_\epsilon$ satisfies

$$N = |\mathcal{B}_N| + |\mathcal{I}_N| \leq \left(81 + \frac{6}{\log 6}\right) \sqrt{M} \Delta_f \epsilon^{-3/2} + \frac{\log(a_0/\epsilon)}{\log 6}. \quad (2.12)$$

Denote the right-hand side of (2.12) by C_ϵ^s . If $T_\epsilon = \infty$, then the inequality holds for every $N \in \mathbb{N}_0$; choosing $N = \lfloor C_\epsilon^s \rfloor + 1$ gives a contradiction. Thus $T_\epsilon < \infty$. For $T_\epsilon \geq 1$, apply (2.12) with $N = T_\epsilon - 1$ and add the final step. This proves (2.3); the case $T_\epsilon = 0$ is immediate. $\square$

3 Adaptive clipped universal trust-region method

Let $\rho_{\min} \leq \rho_0$, $\gamma_+ > 1$, $\gamma_- > 1$, and $0 < \eta \leq 1/64$. At an outer iterate x_k , define

$$g_k := \nabla f(x_k), \quad H_k := \nabla^2 f(x_k), \quad a_k := \|g_k\|, \quad h_k := \max\{a_k, \epsilon\}. \quad (3.1)$$

For a trial value $\rho > 0$, consider

$$\begin{aligned} \min_{d \in \mathbb{R}^n} \quad & g_k^T d + \frac{1}{2} d^T (H_k + \rho \sqrt{h_k} I) d, \\ \text{s. t.} \quad & \|d\| \leq \frac{\sqrt{h_k}}{4\rho}. \end{aligned} \quad (3.2)$$

The subproblem is solved globally. Its trust-region multiplier is denoted by $\lambda \geq 0$.

Lemma 3.1 (Trust-region optimality conditions). *Let (d, λ) be a global primal-dual solution of (3.2). Then*

$$\|d\| \leq \frac{\sqrt{h_k}}{4\rho}, \quad (3.3a)$$

$$\lambda \left(\|d\| - \frac{\sqrt{h_k}}{4\rho} \right) = 0, \quad (3.3b)$$

$$(H_k + \rho \sqrt{h_k} I + \lambda I) d = -g_k, \quad (3.3c)$$

$$H_k + \rho \sqrt{h_k} I + \lambda I \succeq 0. \quad (3.3d)$$

In particular, $\lambda > 0$ implies $\|d\| = \sqrt{h_k}/(4\rho)$.

Given the trial step d , set

$$f_k^+ := f(x_k + d), \quad a_k^+ := \|\nabla f(x_k + d)\|.$$

The trial is accepted if one of the following mutually exclusive tests holds:

$$\lambda = 0 : \quad f_k^+ \leq f(x_k), \quad a_k^+ \leq \frac{1}{2} h_k; \quad (3.4a)$$

$$\lambda > 0 : \quad f(x_k) - f_k^+ \geq \eta \left(\frac{h_k^{3/2}}{\rho} + \lambda \|d\|^2 \right), \quad (3.4b)$$

$$a_k^+ \leq \frac{1}{2} h_k + \lambda \|d\|. \quad (3.4c)$$

Algorithm 1: Adaptive clipped universal trust-region method

Input: $x_0, \epsilon > 0, \rho_0 \geq \rho_{\min} > 0, \gamma_+ > 1, \gamma_- > 1, 0 < \eta \leq 1/64$

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1  $k \leftarrow 0, \rho \leftarrow \rho_0$ 
2 while true do
3   Compute  $g_k, H_k, a_k, h_k$  from (3.1) and  $\mu_k \leftarrow \lambda_{\min}(H_k)$ 
4   repeat
5     if  $a_k \leq \epsilon$  and  $\mu_k \geq -\rho\sqrt{\epsilon}$  then
6       return  $x_k$ 
7     end
8     Solve (3.2) and obtain  $(d, \lambda)$ 
9     Compute  $f_k^+$  and  $a_k^+$ 
10    if (3.4) holds then
11       $x_{k+1} \leftarrow x_k + d$ 
12       $\rho \leftarrow \max\{\rho_{\min}, \rho/\gamma_-\}$ 
13       $k \leftarrow k + 1$ 
14      accept the step
15    else
16       $\rho \leftarrow \gamma_+\rho$ 
17    end
18  until a step is accepted
19 end

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The stopping test in Algorithm 1 is evaluated for every trial value of ρ , including immediately after each rejected trial increases ρ .

4 Trial-step estimates

Lemma 4.1 (Guaranteed trial outcome). *Suppose Assumption 1.1 holds. If*

$$\rho^2 \geq \frac{M}{6}, \quad (4.1)$$

then a trial of Algorithm 1 either satisfies the stopping test or satisfies the acceptance test (3.4).

Proof. If the stopping test holds, the algorithm returns before solving a subproblem. Otherwise it solves (3.2). Let $g = g_k, H = H_k$, and $h = h_k$. By (1.3b), (3.3c), and (3.3d),

$$\begin{aligned} f(x_k + d) - f(x_k) &\leq g^T d + \frac{1}{2} d^T H d + \frac{M}{6} \|d\|^3 \\ &\leq -\frac{1}{2} (\rho\sqrt{h} + \lambda) \|d\|^2 + \frac{M}{6} \|d\|^3. \end{aligned} \quad (4.2)$$

Using (3.3a) and (4.1) gives

$$f(x_k) - f(x_k + d) \geq \frac{1}{4} \rho\sqrt{h} \|d\|^2 + \frac{1}{2} \lambda \|d\|^2. \quad (4.3)$$

Moreover, (1.3a) and (3.3c) imply

$$\begin{aligned}
\|\nabla f(x_k + d)\| &\leq \rho\sqrt{h}\|d\| + \lambda\|d\| + \frac{M}{2}\|d\|^2 \\
&\leq \left(\frac{1}{4} + \frac{M}{32\rho^2}\right)h + \lambda\|d\| \\
&\leq \frac{7}{16}h + \lambda\|d\| \leq \frac{1}{2}h + \lambda\|d\|.
\end{aligned} \tag{4.4}$$

If $\lambda = 0$, (4.3) and (4.4) give (3.4a). If $\lambda > 0$, Lemma 3.1 and (4.3) give

$$f(x_k) - f(x_k + d) \geq \frac{h^{3/2}}{64\rho} + \frac{1}{2}\lambda\|d\|^2,$$

which, together with $\eta \leq 1/64$ and (4.4), gives (3.4b)–(3.4c). Thus every subproblem actually solved with $\rho^2 \geq M/6$ satisfies the appropriate branch of the acceptance test, which proves the stated alternative. $\square$

Corollary 4.1 (Uniform parameter bound). *Every trial parameter generated by Algorithm 1 satisfies*

$$\rho \leq \bar{\rho} := \max \left\{ \rho_0, \gamma_+ \sqrt{\frac{M}{6}} \right\}. \tag{4.5}$$

Every inner loop terminates after finitely many trials.

Proof. Set $\rho_* := \sqrt{M/6}$. By Lemma 4.1, a trial with $\rho \geq \rho_*$ cannot be rejected. Hence, if an inner loop starts with $s > 0$, it stops or accepts no later than the trial with parameter $s\gamma_+^{u_*}$, where

$$u_* := \max \left\{ 0, \left\lceil \frac{\log(\rho_*/s)}{\log \gamma_+} \right\rceil \right\}.$$

Thus every inner loop is finite. Moreover, a rejected parameter satisfies $\rho < \rho_*$, so its successor is smaller than $\gamma_+\rho_*$. After an accepted step, the next initial value is at most the accepted value or $\rho_{\min} \leq \rho_0$. Induction from ρ_0 proves (4.5). $\square$

5 Accepted-step complexity

By Corollary 4.1, each inner loop either returns or accepts a step after finitely many trials. With $\inf \emptyset = \infty$, define the accepted-step stopping index

$$K_\epsilon := \inf \{k \in \mathbb{N}_0 : \text{Algorithm 1 returns } x_k\} \in \mathbb{N}_0 \cup \{\infty\}. \tag{5.1}$$

Thus, if $K_\epsilon = \infty$, the algorithm generates infinitely many accepted steps. For every $k < K_\epsilon$, let ρ_k , d_k , and λ_k denote the parameter, step, and multiplier of the accepted trial producing x_{k+1} .

Let

$$\Delta_f := f(x_0) - f^*, \quad A_\epsilon := \frac{\bar{\rho}\Delta_f}{\epsilon^{3/2}}. \tag{5.2}$$

For any $N \in \mathbb{N}_0$ with $N \leq K_\epsilon$, define the prefix sets

$$\mathcal{B}_N := \{0 \leq k < N : \lambda_k > 0\}, \quad \mathcal{I}_N := \{0 \leq k < N : \lambda_k = 0\}, \quad (5.3)$$

and, for $k \in \mathcal{B}_N$,

$$z_k := \frac{\lambda_k \|d_k\|}{h_k}. \quad (5.4)$$

Lemma 5.1 (Boundary steps). *For every $N \in \mathbb{N}_0$ with $N \leq K_\epsilon$,*

$$|\mathcal{B}_N| \leq \frac{A_\epsilon}{\eta}, \quad \sum_{k \in \mathcal{B}_N} z_k \leq \frac{4A_\epsilon}{\eta}. \quad (5.5)$$

Proof. All accepted steps are monotone. For $k \in \mathcal{B}_N$, (3.4b), $h_k \geq \epsilon$, and $\rho_k \leq \bar{\rho}$ yield

$$f(x_k) - f(x_{k+1}) \geq \frac{\eta \epsilon^{3/2}}{\bar{\rho}}.$$

Summation proves the first bound. By complementarity,

$$\lambda_k \|d_k\|^2 = z_k h_k \|d_k\| = \frac{z_k h_k^{3/2}}{4\rho_k} \geq \frac{z_k \epsilon^{3/2}}{4\bar{\rho}}.$$

Summing the multiplier term in (3.4b) proves the second bound. $\square$

Lemma 5.2 (Gradient-contraction steps). *For every $N \in \mathbb{N}_0$ with $N \leq K_\epsilon$,*

$$|\mathcal{I}_N| \leq |\mathcal{B}_N| + 1 + \frac{\log_+(\|\nabla f(x_0)\|/\epsilon) + \sum_{k \in \mathcal{B}_N} z_k}{\log 2}. \quad (5.6)$$

Proof. Set

$$\Phi_k := \log \frac{h_k}{\epsilon} \geq 0.$$

In particular, $\Phi_0 = \log_+(\|\nabla f(x_0)\|/\epsilon)$. For a boundary step $k \in \mathcal{B}_N$, (3.4c) gives

$$\frac{h_{k+1}}{h_k} \leq \max \left\{ 1, \frac{1}{2} + z_k \right\},$$

and therefore

$$[\Phi_{k+1} - \Phi_k]_+ \leq z_k. \quad (5.7)$$

Partition the interior prefix into

$$\mathcal{F}_N := \{k \in \mathcal{I}_N : h_{k+1} = \epsilon\}, \quad \mathcal{J}_N := \mathcal{I}_N \setminus \mathcal{F}_N.$$

For $k \in \mathcal{J}_N$, (3.4a) gives $\Phi_{k+1} - \Phi_k \leq -\log 2$. If $k \in \mathcal{F}_N$ and $k+1 < N$, then the inner loop at x_{k+1} accepts a step. At each of its trials the stopping test fails and $h_{k+1} = \epsilon$; hence

$$\lambda_{\min}(H_{k+1} + \rho\sqrt{\epsilon}I) < 0,$$

so (3.3d) excludes $\lambda_{k+1} = 0$. Therefore $k + 1 \in \mathcal{B}_N$. The map $k \mapsto k + 1$ is injective, and only $k = N - 1$ can remain unpaired. Thus

$$|\mathcal{F}_N| \leq |\mathcal{B}_N| + 1. \quad (5.8)$$

For $k \in \mathcal{F}_N$, one also has $\Phi_{k+1} - \Phi_k \leq 0$. Telescoping from 0 to N , using $\Phi_N \geq 0$ and (5.7), gives

$$|\mathcal{J}_N| \log 2 \leq \Phi_0 + \sum_{k \in \mathcal{B}_N} z_k.$$

Combining this estimate with (5.8) proves (5.6). $\square$

Theorem 5.1 (Accepted-step complexity). *The stopping index (5.1) is finite and satisfies*

$$K_\epsilon \leq \frac{1}{\eta} \left(2 + \frac{4}{\log 2} \right) A_\epsilon + \frac{\log_+(\|\nabla f(x_0)\|/\epsilon)}{\log 2} + 1 \quad (5.9)$$

Proof. Let C_ϵ^a denote the right-hand side of (5.9). For every $N \in \mathbb{N}_0$ with $N \leq K_\epsilon$, Lemmas 5.1 and 5.2 give

$$N = |\mathcal{B}_N| + |\mathcal{I}_N| \leq \frac{1}{\eta} \left(2 + \frac{4}{\log 2} \right) A_\epsilon + \frac{\log_+(\|\nabla f(x_0)\|/\epsilon)}{\log 2} + 1.$$

If $K_\epsilon = \infty$, this bound holds for every $N \in \mathbb{N}_0$; choosing $N = \lfloor C_\epsilon^a \rfloor + 1$ gives a contradiction. Hence $K_\epsilon < \infty$; taking $N = K_\epsilon$ gives (5.9). $\square$

6 Total complexity and second-order guarantee

Lemma 6.1 (Rejected trials). *Let U_ϵ be the total number of rejected trial steps before termination. Then*

$$U_\epsilon \leq \frac{\log \gamma_-}{\log \gamma_+} K_\epsilon + \frac{\log(\bar{\rho}/\rho_0)}{\log \gamma_+}. \quad (6.1)$$

Proof. Write $K := K_\epsilon < \infty$. For $0 \leq k \leq K$, let s_k be the initial parameter of outer iteration k . For $0 \leq k < K$, let u_k be its number of rejected trials and let $p_k = s_k \gamma_+^{u_k}$ be its accepted parameter; sums over this range are zero when $K = 0$. The update rule gives

$$s_{k+1} \geq \frac{p_k}{\gamma_-},$$

and hence

$$u_k \log \gamma_+ \leq \log s_{k+1} - \log s_k + \log \gamma_-.$$

Summing over $0 \leq k < K$ and using $s_0 = \rho_0$ yields

$$\sum_{k=0}^{K-1} u_k \log \gamma_+ \leq \log \frac{s_K}{\rho_0} + K \log \gamma_-. \quad (6.2)$$

Let u_K be the number of rejected trials in the terminal inner loop at x_K , and let $r_K := s_K \gamma_+^{u_K}$ be the parameter at which the stopping test returns x_K . This includes $u_K = 0$. By [Corollary 4.1](#), $r_K \leq \bar{\rho}$, and hence

$$u_K \log \gamma_+ = \log \frac{r_K}{s_K} \leq \log \frac{\bar{\rho}}{s_K}.$$

Since $U_\epsilon = \sum_{k=0}^{K-1} u_k + u_K$, adding this inequality to [\(6.2\)](#) proves [\(6.1\)](#). $\square$

Theorem 6.1 (Total complexity). *Let Q_ϵ denote the total number of solved trial subproblems. Then*

$$Q_\epsilon \leq \left(1 + \frac{\log \gamma_-}{\log \gamma_+}\right) K_\epsilon + \frac{\log(\bar{\rho}/\rho_0)}{\log \gamma_+}, \quad (6.3)$$

where K_ϵ is bounded in [\(5.9\)](#). The returned point $\hat{x}$ satisfies

$$\|\nabla f(\hat{x})\| \leq \epsilon, \quad \lambda_{\min}(\nabla^2 f(\hat{x})) \geq -\bar{\rho}\sqrt{\epsilon}. \quad (6.4)$$

Proof. The trial bound follows from $Q_\epsilon = K_\epsilon + U_\epsilon$ —the final stopping test solves no subproblem—and [Lemma 6.1](#). At termination, the current parameter is at most $\bar{\rho}$ by [Corollary 4.1](#); the stopping test therefore implies [\(6.4\)](#). $\square$

Corollary 6.1 (Additive overhead). *If $\rho_0 \leq c\sqrt{M}$ for a constant $c > 0$, then*

$$Q_\epsilon = O\left(\sqrt{M} \Delta_f \epsilon^{-3/2} + \log_+(\|\nabla f(x_0)\|/\epsilon) + \log_+(\sqrt{M}/\rho_0) + 1\right). \quad (6.5)$$

In particular, neither the gradient-contraction cost nor the adaptive search cost multiplies the leading $\sqrt{M} \Delta_f \epsilon^{-3/2}$ term.

The Hessian is evaluated once per outer iteration. Thus its evaluation count is at most $K_\epsilon + 1$; reusing accepted trial values, the function and gradient evaluation counts are at most $Q_\epsilon + 1$. The persistent parameter update in [Algorithm 1](#) is the same amortization mechanism used in adaptive regularization methods [\[2\]](#); the multiplier-aware acceptance test is what couples gradient growth to objective decrease.

References

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- [2] Coralía Cartis, Nicholas I. M. Gould, and Philippe L. Toint. Adaptive cubic regularisation methods for unconstrained optimization. part ii: Worst-case function- and derivative-evaluation complexity. *Mathematical Programming*, 130(2):295–319, 2011.